

---

# MEGHANA GUNDA

---

◆ 347-605-0798 ◆ meghanasudarshan24@gmail.com [www.linkedin.com/in/meghana-gunda-35211b184](https://www.linkedin.com/in/meghana-gunda-35211b184)

---

## PROFESSIONAL SUMMARY

---

- An IT professional with 10 years of extensive experience as a Data Engineer and Big Data Developer, specializing in the Big Data Ecosystem for Data Ingestion, Modeling, Analysis, Integration, and Processing.
- Proven expertise in Snowflake, dbt, PySpark, and Kafka, with hands-on experience in large-scale data migrations, including modernizing legacy systems to Snowflake.
- Hands-on experience building and optimizing PySpark applications for both batch and streaming processes, utilizing transformations, Spark SQL queries, and advanced data handling with Data Frames and Datasets.
- Proficient in writing, troubleshooting, and optimizing Spark scripts in Python, supporting robust data workflows and distributed data processing.
- Skilled in utilizing Kafka-Python for distributed messaging, managing both real-time and batch data ingestion and processing with high reliability.
- Extensive knowledge of performance tuning in data warehousing, with experience in techniques such as partitioning, bucketing, and optimizing query performance for high-volume data processing.
- Expertise in real-time analytics using PySpark Streaming and Kafka-Python, along with batch processing workflows on AWS EMR.
- Proven ability to build scalable NoSQL solutions with databases like Cassandra and DynamoDB, integrating them with distributed file systems for efficient data storage and retrieval.
- Deep experience in ETL processes, including data transformation, sourcing, mapping, and loading from diverse sources such as flat files, Excel, Oracle, Teradata, and MSSQL Server, with proficiency in Python-based frameworks.
- Skilled in building production ETL pipelines using Apache Airflow, AWS Glue, and Informatica PowerCenter, automating complex workflows and ensuring data accuracy.
- Strong understanding of data warehousing concepts, including Star Schema, Snowflake Schema, DataMart's, and Kimball Methodology for both relational and multidimensional modeling.
- Experienced with AWS IAM for security compliance, leveraging Python-based tools for secure data access and authentication.
- Knowledgeable in handling diverse file formats, including Avro, Parquet, JSON, and flat files, to support data processing and storage needs.
- Proficient in Python for automation, data processing, and REST API integration, including API management with Flask and Fast API.
- Hands-on experience with Agile/Scrum methodologies for project management, skilled in using JIRA and Confluence for effective collaboration and communication.
- Experienced in data visualization with Python libraries (Matplotlib, Seaborn, Plotly) and integration with Tableau and Power BI for dashboarding and ad-hoc reporting.
- Proficient in working with Kubernetes for containerized and scalable deployment of data processing tasks, ensuring efficient resource management for distributed applications.

---

## SKILLS

---

**Programming Languages:** Python, SQL, C/C++, Shell Scripting

**Big Data Stack:** PySpark, Spark SQL, Kafka (via Kafka-Python), Apache Airflow

**Cloud Platform:** Amazon AWS (EC2, S3, RDS, EMR, Redshift, Glue, Lambda, Athena), Azure (ADF, Synapse, Blob Storage, Logic Apps, Databricks, Key Vault, Monitor, ARM).

**Relational databases:** Oracle, MySQL, SQL Server, PostgreSQL, Teradata, Snowflake

**NoSQL databases:** MongoDB, Cassandra

**Version Control Systems:** Bitbucket, GIT, GitHub

**Operating Systems:** UNIX, Linux, Windows

---

## EDUCATION

---

**Master's:** Information Systems Management

**Robert Morris University** - Moon Township, PA

---

## WORK HISTORY

---

**Senior Data Engineer**

04/2024 - Current

**Nielsen** – Chicago, IL

- Participated in requirement grooming meetings to understand functional requirements from a business perspective and provided estimates for converting requirements into software solutions. Designed, developed, and delivered code to IT/UAT/PROD, validating and managing data pipelines from multiple applications using Agile development with sprints in JIRA.
- Responsible for data validation and monitoring in Cassandra tables and EC2 instance status checks across DEV, QA, CERT, and PROD environments in AWS.
- Conducted analysis on existing data flows, creating high-level and low-level technical design documents for business stakeholders to ensure technical designs align with business requirements.
- Created and deployed PySpark jobs across different environments, loading data into the NoSQL database Cassandra. Secured data through encryption techniques for sensitive data.
- Integrated Databricks for advanced analytics alongside AWS Lambda and Kinesis Firehose.
- Designed and implemented scalable data pipelines to support the migration of legacy systems into Snowflake while optimizing for performance and cost-efficiency.
- Implemented AWS solutions with EC2, S3, RDS, EBS, Elastic Load Balancer, Auto Scaling Groups, and CloudWatch for optimized monitoring. Configured monitors, alarms, and notifications in CloudWatch to track EC2 performance.
- Developed code using PySpark for distributed data processing, managed data storage in Cassandra, and used Kafka-Python for real-time data streaming. Employed Kubernetes for containerized application deployments, Docker pipelines for CI/CD, and GitHub for version control.
- Scheduled and monitored data workflows in Apache Airflow as an alternative to Autosys, providing better flexibility in managing ETL job dependencies and scheduling.
- Extensively worked on data pipeline design with Informatica PowerCenter, creating mappings, workflows, variables, and session parameters to streamline ETL processes.

- Benchmarked and optimized data load pipelines to meet established performance standards, ensuring efficient and reliable data delivery for downstream applications.
- Validated data flow between source and target using Python-based testing scripts to identify and troubleshoot invalid mappings within Informatica jobs.
- Orchestrated ELT workflows with DBT for Snowflake, improving transformation logic and audibility.
- Utilized SQL tools like SQL Developer to execute queries and validate data for accuracy and consistency across ETL processes.
- Studied existing systems, conducted code and job reviews, and provided comprehensive feedback for optimization, enabling continuous improvement across data pipelines.
- Coordinated with offshore teams to ensure timely and quality deliverables, overseeing production deployments across DEV, QA, CERT, and PROD environments.
- Conducted testing and performance tuning in AWS RDS and Redshift databases using complex SQL scripts and query optimization techniques to manage performance effectively.

## **Senior Data Engineer.**

08/2023 - 03/2024

### **Honeywell – Atlanta, GA**

- As a Data Engineer, I am responsible for building scalable, distributed data solutions using Apache PySpark and Databricks, optimizing data transformations and streamlining data engineering processes.
- Involved in Agile Development process, including Scrum and Sprint planning.
- Handled PySpark cluster installations in a Linux environment on AWS.
- Migrated on-premise environment to AWS.
- Migrated data warehouses to Amazon Redshift and S3 Data Lake.
- Migrated large-scale datasets from on-premise systems to Snowflake and automated data transformations with DBT.
- Defined virtual warehouse sizing and resource allocation for Redshift clusters to handle different workload types.
- Involved in porting existing on-premise SQL code migration to Amazon Redshift and Athena for scalable querying.
- Involved in migration of an Oracle SQL ETL to run on AWS using EMR, Glue, and Redshift with CloudWatch for job monitoring and scheduling with Apache Airflow.
- Extracted data from data lakes and EDWs to relational databases for analysis and insights using SQL queries and PySpark.
- Developed PySpark scripts to merge static and dynamic files and cleanse the data.
- Created PySpark procedures, functions, and packages for data loading processes.
- Integrated ML-based fraud detection using PySpark Streaming and Databricks.
- Designed, developed, and maintained data integration programs in a PySpark and RDBMS environment with both traditional and non-traditional source systems.
- Developed PySpark jobs to parse raw data, populate staging tables, and store refined data in partitioned tables within the data lake.
- Designed and deployed stream sets pipelines for ingesting data into Snowflake from legacy relational systems.
- Wrote PySpark scripts as an alternative to Sqoop for data movement between relational databases and HDFS/S3.
- Set up a Data Lake in AWS using S3 and Athena for scalable storage and querying.
- Developed scripts in Athena to connect to reporting tools for analytics and reporting.
- Designed workflows using Apache Airflow to automate data transformation and change data capture processes.
- Carried out data transformation and cleansing using SQL queries and PySpark.
- Used Kafka-Python and Spark Streaming to ingest real-time or near-real-time data into HDFS/S3.
- Worked on PySpark APIs for data transformations.

- Automated workflows in Apache Airflow to streamline ELT processes in Snowflake.
- Built monitoring reports for data loads into AWS data lakes, ensuring site-level data reliability.
- Participated in daily stand-ups, bi-weekly scrums, and PI planning

## Senior Data Engineer

10/2020 - 11/2022

### Accenture – India

- Extensive experience in working with AWS cloud platform (EC2, S3, EMR, Redshift, Lambda, and Glue).
- Proficient in PySpark for data processing, including DataFrame API, Spark SQL, and PySpark Streaming.
- Developed PySpark applications to handle data ingestion from various RDBMS and streaming sources, optimizing for high performance and scalability.
- Improved performance and optimized existing data processing algorithms by replacing Hadoop components with PySpark on AWS.
- Used Spark Context, Spark SQL, and DataFrame API to perform transformations and in-memory computations.
- Configured Databricks Clusters for high-performance data processing and supported on-demand data exploration, which enabled the team to scale processing for large datasets. Implemented parallel processing using PySpark and Databricks.
- Implemented PySpark Streaming to process real-time data pipelines and capture incoming data from Kafka.
- Developed a Kafka consumer API in Python to ingest data from Kafka topics and process it using PySpark.
- Processed XML messages from Kafka, flattening the data into JSON files using PySpark Streaming to capture UI updates.
- Created PySpark jobs to preprocess and flatten JSON documents into flat files for downstream processes.
- Transitioned Oracle SQL and Teradata datasets to Snowflake, optimizing query performance and storage utilization.
- Leveraged in-memory data computation with PySpark to generate real-time responses from D-Stream data.
- Migrated an on-premises application to AWS, utilizing EC2 and S3 for data processing and storage.
- Maintained a distributed data processing environment on AWS EMR with PySpark.
- Loaded data into S3 buckets using AWS Glue and PySpark, and filtered data stored in S3 with Elasticsearch.
- Configured AWS Glue for automated data integration, filtering, and transformation, loading data into Redshift for analysis.
- Designed and managed column families in Cassandra to store ingested data and perform business-required transformations.
- Utilized the Spark Cassandra Connector for data loading, supporting read and write operations between PySpark and Cassandra.
- Created data models for transactional logs and executed Cassandra Query Language (CQL) to analyze data for business insights.
- Performed stress tests on Cassandra using Cassandra-stress tool to improve read and write performance.
- Used AWS Athena and Redshift to query, analyze, and aggregate data for reporting and downstream applications.
- Ingested real-time data from Apache Kafka for data analysis and engineering, securing and optimizing Kafka for performance.
- Experienced with data formats including Avro, Parquet, JSON, and developed Python UDFs for sorting and transformation.
- Developed Python-based Custom UDFs for Hive, enabling flexible data manipulation and preparation for analysis.
- Scheduled and automated data pipelines using Apache Airflow as a replacement for Oozie and Autosys for ETL orchestration.

- Replaced Sqoop and MapReduce jobs with PySpark scripts for efficient data ingestion from RDBMS and external systems.
- Developed CI/CD pipelines using Jenkins for continuous integration and deployment with PySpark applications.
- Set up Kerberos authentication for secure network communication across distributed nodes.
- Managed and monitored clusters on AWS using CloudWatch and eliminated the need for Cloudera Manager.

## **Data Engineer**

02/2018 - 09/2020

### **Cognizant – India**

- Worked as a Sr. Data Engineer using Python-based tools with data ecosystems such as Cassandra, PySpark, Apache Kafka, and AWS.
- Involved in Agile development methodology as an active member in Scrum meetings.
- Designed and implemented scalable Cloud Data and Analytical architecture solutions using AWS services, including Redshift, S3, Lambda, Glue, and EMR. Involved in end-to-end development of data pipelines using PySpark, Python scripts, Apache Kafka, and Apache Airflow.
- Implemented AWS services such as Redshift, RDS, Athena, Glue, and S3 for data warehousing, transformation, and storage solutions.
- Extracted and loaded data into AWS S3 Data Lake environments using PySpark, accessible for business analysis and reporting.
- Managed and supported data warehousing operations, advanced predictive applications, and big data solutions using AWS Glue and Redshift.
- Developed PySpark scripts to transform raw data into structured datasets as specified by business users.
- Utilized PySpark and Python to develop and execute Big Data analytics and machine learning applications, handling use cases with Spark MLlib.
- Installed and configured HDFS, PySpark, Kafka, and Apache Airflow to develop and manage data pipelines for data cleansing and pre-processing.
- Used AWS Glue Data Catalog to perform data transformations and enable fast querying in Athena and Redshift.
- Improved performance and optimized data processing algorithms using PySpark DataFrames, Spark SQL, and AWS EMR.
- Developed PySpark code for faster testing and processing of data in a scalable environment on AWS.
- Built data ingestion pipelines from Kafka and processed it in PySpark, storing transformed data in Cassandra.
- Documented requirements, existing code, and Spark data pipelines for data integration, analysis, and machine learning applications.
- Used Apache Kafka for real-time data ingestion, transforming and storing data in HDFS/S3 for analysis.
- Explored machine learning algorithms using MLlib in PySpark for business use cases in predictive analytics.
- Performed data analysis on Parquet tables stored in S3 using AWS Athena and Redshift to meet business requirements.
- Configured Apache Airflow workflows to automate PySpark and Python data processing jobs, ensuring smooth data integration.
- Imported and exported analyzed data to relational databases and Amazon Redshift for visualization and reporting for the BI team.

## **Data Engineer**

08/2015 - 01/2018

### **Avaya – India**

- Developed predictive models using machine learning algorithms like XGBoost, Random Forests, and Logistic Regression to predict credit risk and identify potential fraud cases with high accuracy.

- Utilized Time-Series forecasting algorithms (e.g., ARIMA, SARIMA, Holt-Winters) to forecast metrics such as productivity, revenue, and sales for business planning.
- Conducted large-scale exploratory data analysis and data mining to identify patterns, correlations, and anomalies using AWS Redshift, Glue, and S3.
- Built and maintained complex ETL data pipelines and optimized data flows in AWS for large-scale data processing.
- Created data visualizations and dashboards in Tableau and Python (using Matplotlib and Seaborn) for insights on key metrics and client-specific projects.
- Developed complex data models with Python and SQL to enhance data quality and ensure data reliability in data warehouses, including Redshift and S3.
- Collaborated with cross-functional teams for data quality improvement, data modeling, and optimization of data processing workflows.
- Conducted correlation analysis and visualizations in Python (Matplotlib, Seaborn) to understand customer loan patterns and segmentation.
- Worked with Data Engineering and Data Services teams for optimal ETL processes, leveraging SQL, Python, and AWS-native tools.